Promo Finder: finding the best promos in a microdrama, automatically

Kuku AI Generalist Assignment · Problem 1 · Vaibhav Arya · 231112 · B.Tech Civil Engineering · Prototype: [GitHub link] · Raw outputs: [Drive link] · Video: [link]

1. Why this problem

Promos are how microdrama apps buy every new viewer. Editors currently watch each episode to pick them, and Kuku releases shows weekly, so this is a daily, repeated cost on the business's main growth lever. It also has real data (50 timestamped episodes), an output that can be checked against timestamps, and it matches the interview test (re-run on a new show). Problem 2 had no data and a high risk of invented law; Problem 3 needed proof of the problem first.

2. Assumptions (the brief was silent)

- Script timestamps ≈ final-cut timestamps. Audience: US social users who have never seen the show.
- 'He was a Duke' is in the title, so the premise is not a spoiler; later twists are retargeting-only.
- Visual beats (strikes, explosions) count as hooks, since vertical ads are often watched muted. Multi-part edits keep playback order.
- Graphic violence and sexual content are flagged for typical Meta/TikTok ad policy, not silently dropped. No access to real ad performance, so 'best' = the brief's three tests plus posting strategy.

3. How it works

Parse → score every beat → generate cuts → rank → editor reviews. (1) **Parse** 3,727 timed blocks; camera lines count for time but never decide a cut. (2) **Score each line/action** for hook, cliffhanger, intensity, context-need and safety, using the script's own emotion tags, identity-reveal patterns ('he is marrying me', 'your grace'), physical actions, **overlap with the episode logline** (the writer's own summary of the key beat; this was the biggest single gain), **'cut right before the payoff'**, and penalties for endings that resolve (an embrace, 'Are you hurt?'), even when the script labels them 'cliffhanger'. (3) **Generate ~1,400 cuts**: continuous, 'pruned' (strong hook + strong ending with the weak middle removed, as an editor would) and cross-episode. Every cut is snapped to line boundaries and kept within 20–60 s *by code*, so timestamps are never an AI guess. (4) **Rank** with diversity (one promo per hook moment; the rest become A/B 'alternate cuts'), then a **posting plan**: wave 1 = early free episodes (a new viewer can start now, nothing spoiled), wave 2 = mid-story, wave 3 = late payoffs for retargeting. (5) An editor approves the shortlist in a review page.

Kept as code	Given to AI (optional, one config line)	Human in the loop
Parsing, timing, line-boundary snapping, 20–60 s, dedupe, safety flags, outputs. Free, exact, testable.	Meaning: is this a reveal? does it resolve? would a stranger follow it? Beat tagging (50 calls) + shortlist judge (60 calls), blended with the rule scores. OpenAI-compatible (Gemini/Groq/Ollama/any vendor API) or Anthropic.	Editor approves ~10 picks in review.html (minutes instead of 100 min of footage); checks flagged multi-part edits and ad-policy flags.

Runs offline with no key (rule mode, ~3 s for 50 episodes). With a model, the measured cost is ~150k tokens per show: about $0.03 on a Flash-class model, free within Gemini's free tier, and cached so re-runs are free. Scale: 10 shows a week ≈ 1,100 calls; the editor's review is the bottleneck, not compute.

4. What it found and what I'd post first

Wave	Promo (tool output)	Hook (first 3 s)	Cliffhanger	Why it works
1	E3 01:24.0–01:38.5 + 02:13.2–02:25.8 (27 s)	Scarred stranger drops a guard dragging the bride	He kneels with a ring; her answer withheld	Whole premise, free episode, works muted
1	E12 00:44.0–01:27.2 (43 s)	'I want to see you crawl, sister.'	'Tell me where the garden is' (freeze)	Cruel ultimatum, clear stakes, a decision left hanging
1	E1 00:39.1–01:14.3 (35 s)	'Because he is marrying me.'	'How long?'	Altar betrayal, episode 1, no context needed
2	E24 00:32.5–01:08.0 (36 s)	Riding-crop strike	Crop turned on her friend	The show's only true 'slap' beat
2	E23 01:19.0–02:04.0 (45 s)	'Someone is poisoning her.'	'Someone is coming.'	Reveal, then an interruption at the peak

Full ranked list (30 promos, alternate cuts, 10 cross-episode variants, the posting plan) is in the raw outputs.

5. How I know it's good, and where it fails

Check	Result
Hard rules (independent re-check of 40 outputs)	40/40 within 20–60 s, no line cut at any edge; deterministic across runs
Blind audit of top 30 against the brief's 3 tests	**8/10 of the top 10** and 20/30 overall pass all three (hook 27, cliffhanger 28, standalone 24)
Gold set: 24 promos I picked by reading all 50 eps	Generator finds 22/24 (92%); top-30 recall 0.38; top-10 precision ≥ 0.50
Overfitting guard: tune on eps 1–25, test on 26–50	Kept only changes that held up on test. A 150-trial weight search raised tune recall 0.31→0.54 but **dropped test 0.36→0.27, so I rejected it**.

Where it picks badly:

- **Context-heavy clips** (6/30): names like Sterling or 'the council' mean nothing to a stranger. Rules catch names, not plot.
- **Quiet twists**: 'Your Grace.' said to the 'beggar' husband is the show's core reveal, but rules score it low because it's calm; 'Mother.' slipped in as a hook. Payoffs ('This is my son… the rightful Duke!') can pass for cliffhangers.
- **Cross-episode edits**: only ~4/10 usable without an editor or LLM check, so they're kept out of the main ranking in rule mode.
- **Text ≠ screen**: acting, faces and music are invisible; the gold set and audit come from one rater. Next step: an editor re-rates and the posted promos' 3-second hook rate, completion and CPI are compared with editor-picked promos. That's how we'd know it works in production.

The LLM layer's 'context' and 'resolves' scores target the first two failures, and measuring it is the first thing to do with a key.

6. AI tools used, and where they got it wrong

Claude (Claude Code) for research, script reading, coding and tests. Where it got things wrong:

- Its first list of 24 hand-picked cuts **clipped dialogue mid-line 9 times** ('There it is. Ha!'), which is why cut points are now code, never model output.
- The scripts themselves: the Duke's estate is spelled **11 different ways** (Ashburg, Ashbourne, Ashburton…), a marriage of a few days is called 'three years', and a romantic embrace is labelled 'cliffhanger'. The tool had to stop trusting those labels.
- Its suggested weight tuning **overfit** (above), and tests caught a JSON parser it wrote that misread array replies.

7. Scaling to every show

Same format means no per-show setup: leads, names and aliases are inferred from the scripts. The weekly flow is script → run (seconds) → editor approves in review.html → export. Approve/reject decisions are logged and become new gold labels for re-tuning and per-genre weights. Next steps: add the LLM judge, learn weights from real ad metrics, and generate several A/B cuts per hook.